

# TFM - $\mu y$ Vs. $y$ : TT plots

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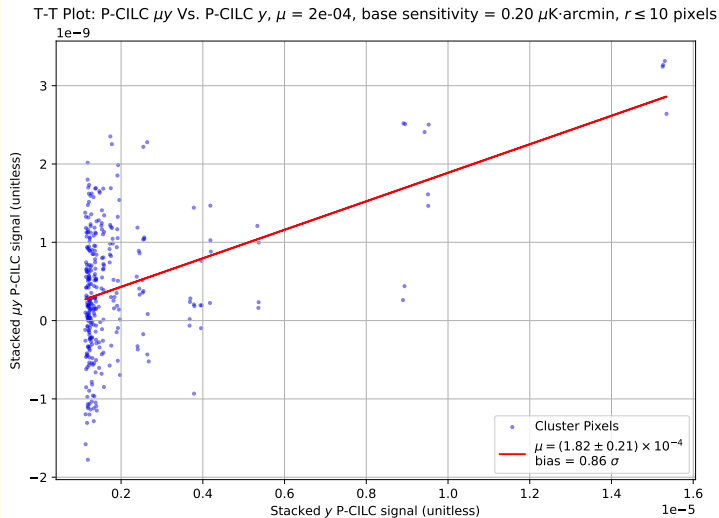
Joan Ribot Oliver

2026

Universidad de Cantabria

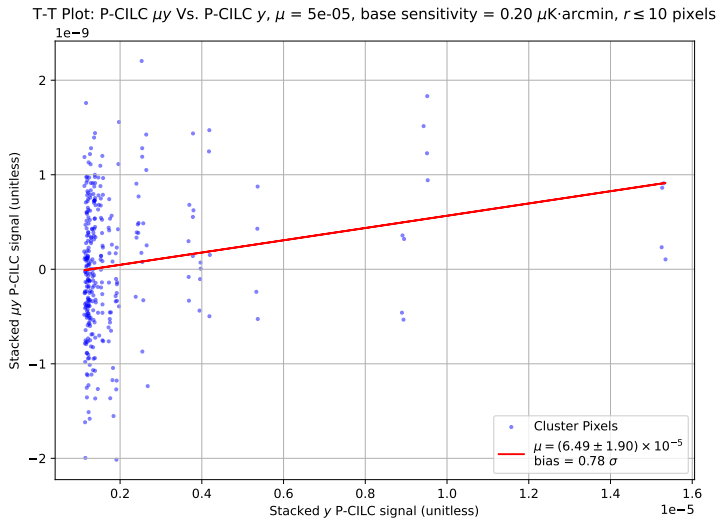
## TT plot: P-CILC

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu\text{K}\cdot\text{arcmin}$ ), s.m. =  $10^4$   
 $\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



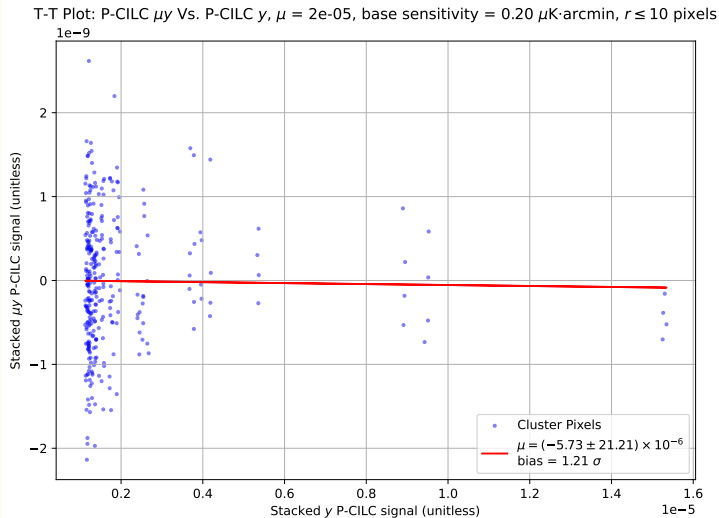
# TT plot: P-CILC

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ ), s.m. =  $10^4$   
 $\mu = 5 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



# TT plot: P-CILC

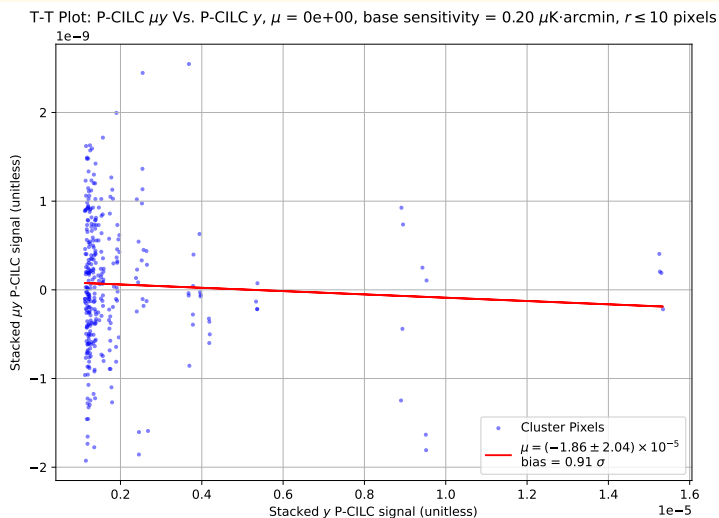
4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ ), s.m. =  $10^4$   
 $\mu = 2 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



# TT plot: P-CILC

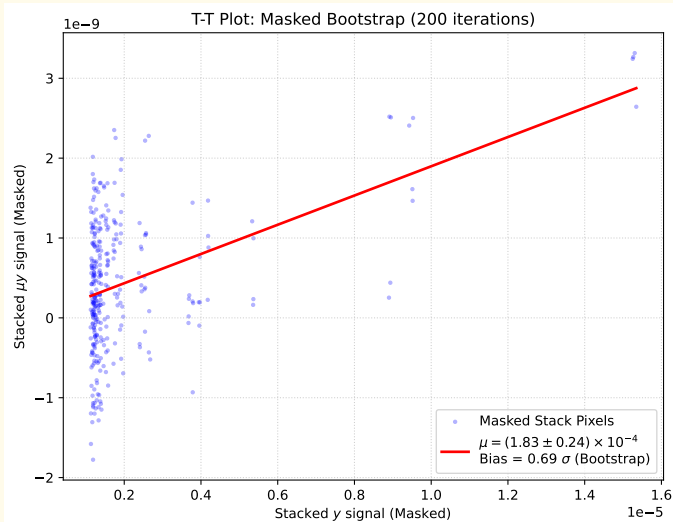
3 layers: tSZ, CMB and noise ( $0.2\mu\text{K}\cdot\text{arcmin}$ ), s.m. =  $10^4$

Null test:  $\mu = 0$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



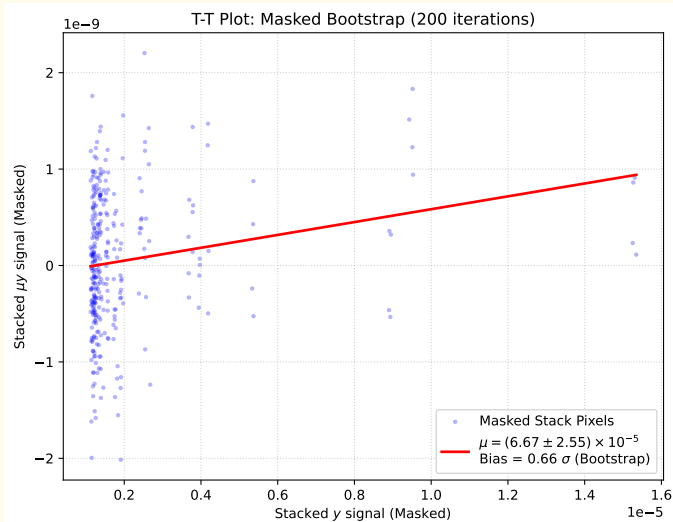
# TT plot: P-CILC + BOOTSTRAP

4 layers: tSZ, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ ), s.m. =  $10^4$ , Bootstrap  
 $\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



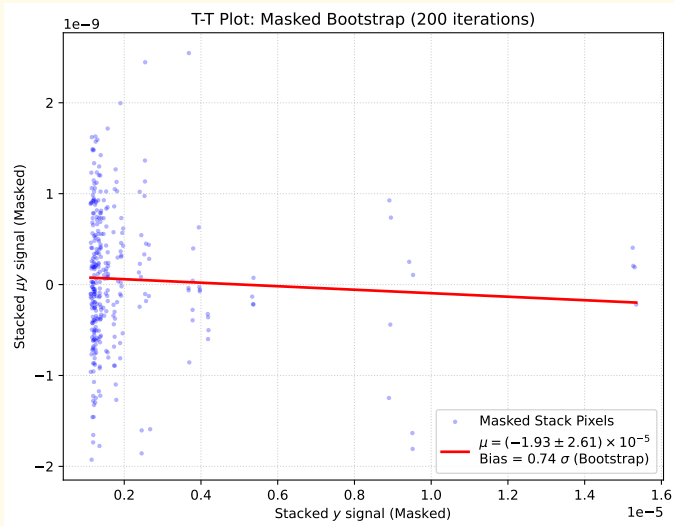
# TT plot: P-CILC + BOOTSTRAP

4 layers: tSZ, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ ), s.m. =  $10^4$ , Bootstrap  
 $\mu = 5 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



# TT plot: P-CILC + BOOTSTRAP

3 layers: tSZ, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ ), s.m. =  $10^4$ , Bootstrap  
Null test:  $\mu = 0$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$

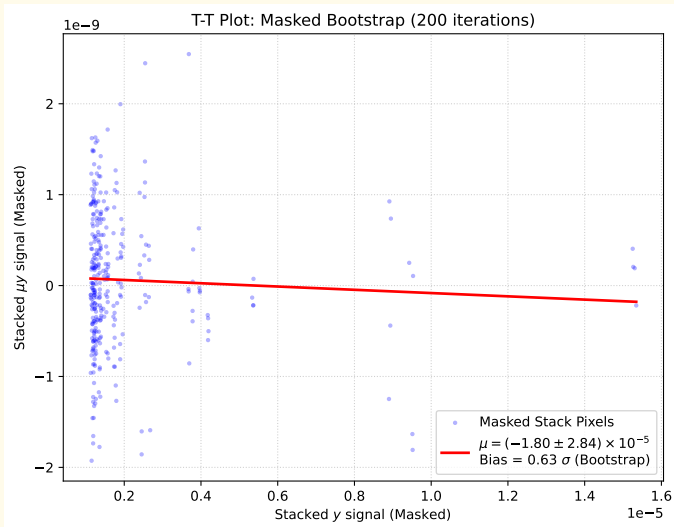




# TT plot: P-CILC + BOOTSTRAP

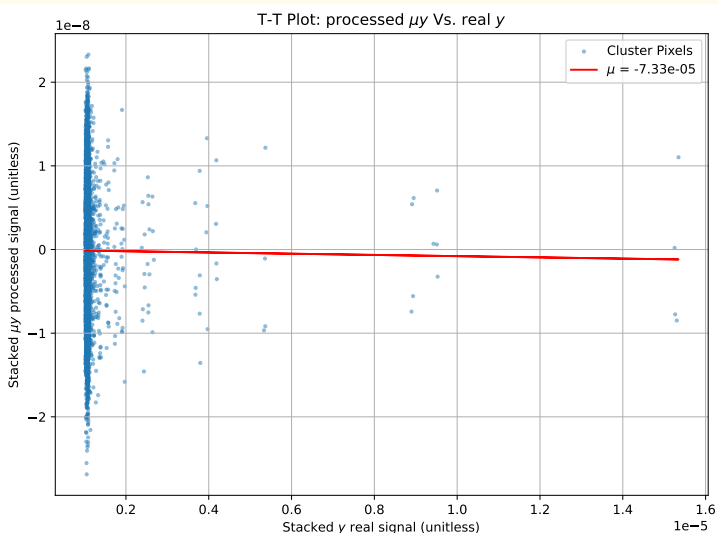
3 layers: tSZ, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ ), s.m. =  $10^4$ , Bootstrap

Null test:  $\mu = 0$ , freqs = *PICO*,  $r = 10$  pixels =  $15.0'$



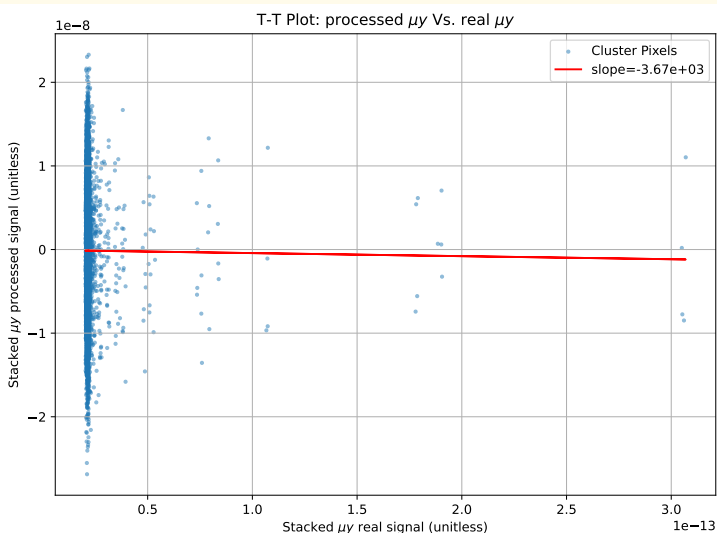
## TT plot: P-ILC no $y$ input

3 layers:  $\mu$ -distortion, CMB and noise ( $30\mu K \cdot \text{arcmin}$ )  
 $\mu = 2 \times 10^{-8}$ , freqs = [30, 40, 90, 150, 220, 270] GHz



## TT plot: P-ILC no $y$ input

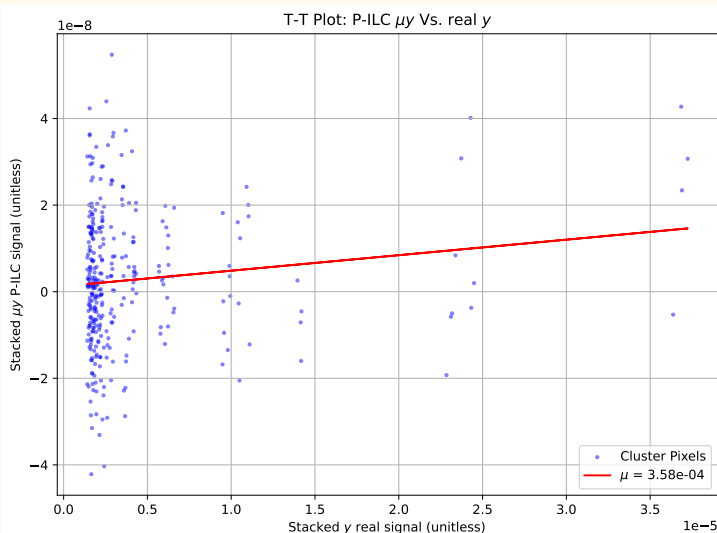
3 layers:  $\mu$ -distortion, CMB and noise ( $30\mu K \cdot \text{arcmin}$ )  
 $\mu = 2 \times 10^{-8}$ , freqs = [30, 40, 90, 150, 220, 270] GHz



## TT plot: P-ILC no $y$ input

**3 layers:**  $\mu$ -distortion, CMB and noise ( $2\mu K \cdot \text{arcmin}$ )

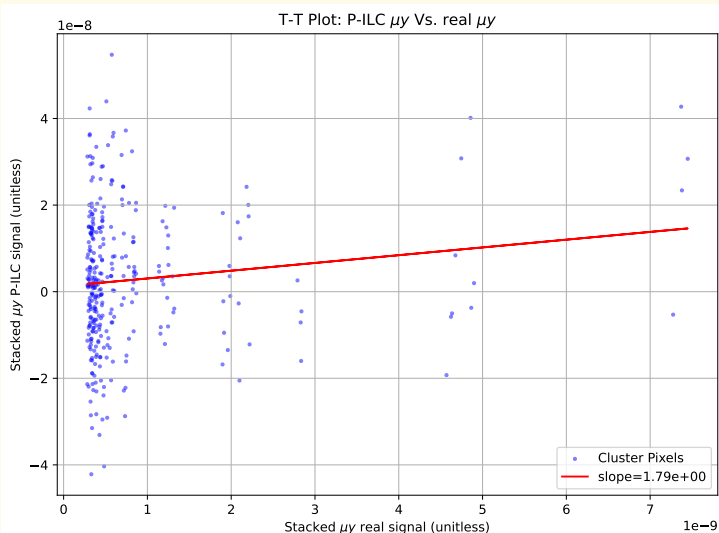
$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



## TT plot: P-ILC no $y$ input

**3 layers:**  $\mu$ -distortion, CMB and noise ( $2\mu K \cdot \text{arcmin}$ )

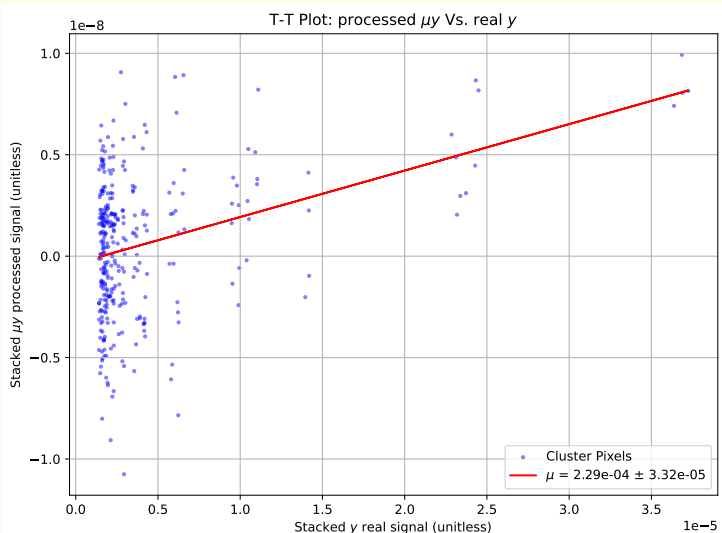
$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



## TT plot: P-ILC no $y$ input

**3 layers:**  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

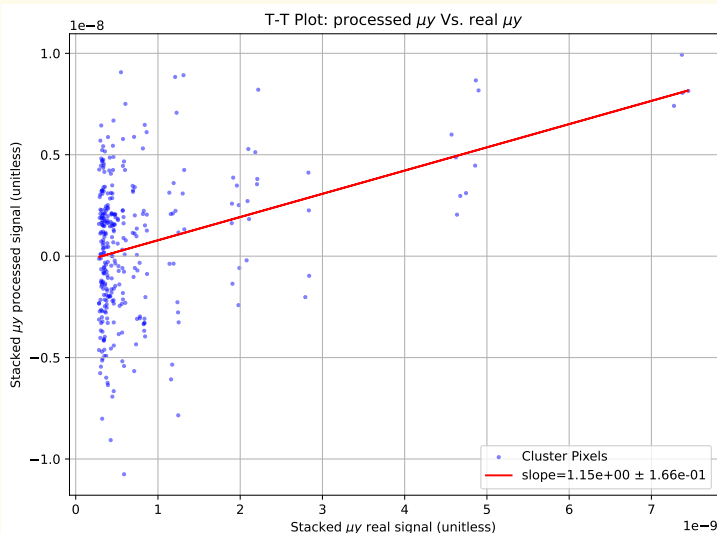
$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



## TT plot: P-ILC no $y$ input

**3 layers:**  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

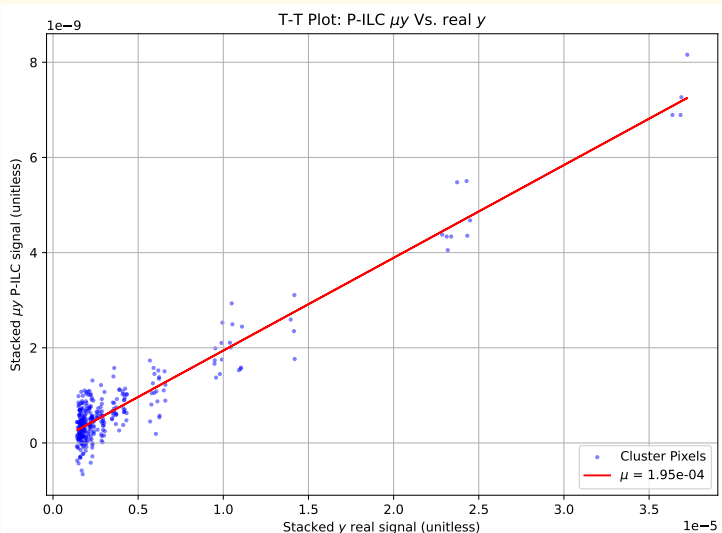
$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$



## TT plot: P-ILC no $y$ input

**3 layers:**  $\mu$ -distortion, CMB and noise ( $0.02\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$





## TT plot: P-ILC no $y$ input

**3 layers:**  $\mu$ -distortion, CMB and noise ( $0.02\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz,  $r = 10$  pixels =  $15.0'$

